

12.18

(a)

```
> p_mother <- mean(mroz_working$mothercoll) * 100
> p_father <- mean(mroz_working$fathercoll) * 100
> cat("母親有大學學歷比例:", p_mother, "%\n")
母親有大學學歷比例: 12.14953 %
> cat("父親有大學學歷比例:", p_father, "%\n")
父親有大學學歷比例: 11.68224 %
```

#

(b)

```
> #b小題
> cor_matrix<-cor(mroz_working[,c("educ","mothercoll","fathercoll")])
> print(cor_matrix)
```

	educ	mothercoll	fathercoll
educ	1.0000000	0.3594705	0.3984962
mothercoll	0.3594705	1.0000000	0.3545709
fathercoll	0.3984962	0.3545709	1.0000000

需要看educ和父母的coll變數的相關係數，如果係數顯著大於 0（通常約 0.2~0.4），則滿足工具變數的「相關性」前提。而dummy變數較好的原因在於門檻效應，父母是否具有大學學歷對子女的影響可能會比多讀一年書更具有決定性，以及減少測量誤差，受訪者對於父母是否有大學學歷通常會比精確讀了幾年更了解。

#

(c)

```
Call:
ivreg(formula = log(wage) ~ exper + I(exper^2) + educ | exper +
      I(exper^2) + mothercoll, data = mroz_working)

Residuals:
      Min       1Q   Median       3Q      Max
-3.08719 -0.32444  0.04147  0.36634  2.35621

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.1327561   0.4965325   -0.267   0.78932
exper         0.0433444   0.0134135    3.231   0.00133 **
I(exper^2)   -0.0008711   0.0004017   -2.169   0.03066 *
educ         0.0760180   0.0394077    1.929   0.05440 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared:  0.147,    Adjusted R-squared:  0.1409
Wald test:      8.2 on 3 and 424 DF,  p-value: 2.569e-05

> cat("EDUC 係數的 95% 信心區間:\n")
EDUC 係數的 95% 信心區間:
> confint(iv_model_c, "educ", level = 0.95)
              2.5 %      97.5 %
educ -0.001219763  0.1532557
```

The 95% C.I for educ is : $[-0.0012, 0.1533]$ #

(d)

```
> f_test_d<-linearHypothesis(first_stage_d,"mothercoll")
> cat("\nmothercoll 的 F 檢定統計量:\n")

mothercoll 的 F 檢定統計量:
> print(f_test_d)

Linear hypothesis test:
mothercoll = 0

Model 1: restricted model
Model 2: educ ~ mothercoll + exper + I(exper^2)

      Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1       425 2219.2
2       424 1929.9  1     289.32 63.563 1.455e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F值為 63.563 > 10, 根據經驗法則, 若 $F > 10$, 可認定此工具變數為強工具變數, 代表 MOTHERCOLL 是個強大的工具變數, 其對 EDUC 有顯著的預測能力。

#

(e)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | mothercoll +
  fathercoll + exper + I(exper^2), data = mroz_working)

Residuals:
    Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819   0.3922213  -0.712   0.47714
educ         0.0878477   0.0307808   2.854   0.00453 **
exper        0.0426761   0.0132950   3.210   0.00143 **
I(exper^2)   -0.0008486   0.0003976  -2.135   0.03337 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,    Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06

> cat("\nEDUC 係數的 95% 信心區間:\n")

EDUC 係數的 95% 信心區間:
> confint(iv_model_e, "educ", level = 0.95)
                2.5 %          97.5 %
educ 0.02751845 0.1481769
```

The 95% CI for EDUC is: $[0.0275, 0.1482]$, 比 (c) 的 CI 更窄。

(f)

```
10.18 (f) 第一階段迴歸結果:
> # 檢定 mothercoll 與 fathercoll 是否「同時」不為 0
> f_test_f <- linearHypothesis(first_stage_f, c("mothercoll = 0", "fathercoll = 0"))
> print(f_test_f)

Linear hypothesis test:
mothercoll = 0
fathercoll = 0

Model 1: restricted model
Model 2: educ ~ mothercoll + fathercoll + exper + I(exper^2)

    Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1      425 2219.2
2      423 1748.3  2      470.88 56.963 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$\therefore F = 56.963 > 10$, 代表此工具變數足夠強大。

(g)

```
Diagnostic tests:
              df1 df2 statistic p-value
Weak instruments  2 423    56.963 <2e-16 ***
Wu-Hausman       1 423     0.519   0.472
Sargan           1  NA     0.238   0.626
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,    Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06
```

Weak instruments的F值為56.96，顯示MOTHERCOLL與FATHERCOLL是強工具變數。

Sargan test的P值為0.626，顯示工具變數滿足外生性要求，是有效的工具。

而Wu-Hausman test的P值為0.472，未能拒絕教育程度為外生的假設，代表在該樣本中，IV估計與OLS估計的差異在統計上並不顯著。

#

10.20

(a)

```

> #a小題
> model_ols<-lm(y~x,data=capm4)
> summary(model_ols)

Call:
lm(formula = y ~ x, data = capm4)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27541 -0.05240 -0.00780  0.04239  0.35082

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.006098   0.007747   0.787   0.433
x            1.318947   0.160790   8.203 1.98e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.089 on 130 degrees of freedom
Multiple R-squared:  0.3411,    Adjusted R-squared:  0.336
F-statistic: 67.29 on 1 and 130 DF,  p-value: 1.979e-13

```

$$r_j - r_f = \alpha_j + \beta_j (R_m - R_f)$$

$\therefore \hat{\beta}_j = 1.318947$ $\therefore$ It is risky relative to the market portfolio. #

(b)

```

Call:
lm(formula = x ~ rank, data = capm4)

Residuals:
    Min       1Q   Median       3Q      Max
-0.106351 -0.006045  0.002346  0.009094  0.017870

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.978e-02  2.732e-03  -29.20  <2e-16 ***
rank         1.197e-03  3.565e-05   33.59  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01561 on 130 degrees of freedom
Multiple R-squared:  0.8967,    Adjusted R-squared:  0.8959
F-statistic: 1128 on 1 and 130 DF,  p-value: < 2.2e-16

```

rank variable 是由 market risk premium 排序得來, 故兩者相關性極強, 而計量學者認為排名能過濾一部分隨機的測量噪音, 且 (a) 小題 β_j 顯著不為 0, 這些情況可認為 rank variable 滿足 IV1-IV3 之條件。

$\therefore p\text{-value} < 2 \times 10^{-16}$ $\therefore$ The coefficient of RANK is very significant.

$\therefore R^2 = 0.9867$ $\therefore$ The RANK can be regarded as a strong IV. #

(c)

```
> #c小題
> v_hat<-resid(first_stage_b)
> model_augmented<-lm(y~x+v_hat,data=capm4)
> summary(model_augmented)

Call:
lm(formula = y ~ x + v_hat, data = capm4)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27249 -0.04573 -0.00961  0.03809  0.33946

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.006116   0.007651   0.799   0.4255
x            1.430572   0.167696   8.531 3.39e-14 ***
v_hat       -1.080197   0.521665  -2.071  0.0404 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0879 on 129 degrees of freedom
Multiple R-squared:  0.3623,    Adjusted R-squared:  0.3524
F-statistic: 36.64 on 2 and 129 DF,  p-value: 2.509e-13
```

$\therefore \uparrow \hat{z}$ $p\text{-value} = 0.0404 > 0.01$ $\therefore$ 在 1% 顯著水準下, 我們無法拒絕市場報酬率為外生的假設。#

(d)

```
Call:
ivreg(formula = y ~ x | rank, data = capm4)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27716 -0.05101 -0.01067  0.04235  0.34684

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.006116   0.007761   0.788   0.432
x            1.430572   0.170117   8.409 6.36e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08917 on 130 degrees of freedom
Multiple R-Squared:  0.3386,    Adjusted R-squared:  0.3335
Wald test: 70.72 on 1 and 130 DF,  p-value: 6.356e-14

> # 比較 OLS 與 IV 的係數
> cat("OLS 估計的 Beta:", coef(model_ols)["x"], "\n")
OLS 估計的 Beta: 1.318947
> cat("IV 估計的 Beta:", coef(model_iv)["x"], "\n")
IV 估計的 Beta: 1.430572
```

在有測量誤差的情況下, OLS 估計出的 Beta 會存在衰減偏誤, 導致 Beta 被低估, 而 IV 估計出的 Beta 大於 OLS 估計出的 Beta, 代表此 IV variable 估計符合我們的預期。#

(e)

```
> #e小題
> capm4$pos <- ifelse(capm4$x > 0, 1, 0)
> first_stage_e <- lm(x ~ rank + pos, data = capm4)
> summary(first_stage_e)

Call:
lm(formula = x ~ rank + pos, data = capm4)

Residuals:
    Min       1Q   Median       3Q      Max
-0.104886 -0.005767  0.002552  0.009066  0.017805

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.135e-02  2.864e-03  -28.404  <2e-16 ***
rank         1.298e-03  6.889e-05   18.841  <2e-16 ***
pos         -9.028e-03  5.300e-03   -1.704   0.0909 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01549 on 129 degrees of freedom
Multiple R-squared:  0.8989,    Adjusted R-squared:  0.8974
F-statistic: 573.7 on 2 and 129 DF,  p-value: < 2.2e-16
```

$\therefore F = 573.7 > 10$ $\therefore$ We can say that we have adequately strong IV.
 $R^2 = 0.8989$

(f)

```
Call:
lm(formula = y ~ x + v_hat_e, data = capm4)

Residuals:
    Min       1Q   Median       3Q      Max
-0.26882 -0.04602 -0.00947  0.03698  0.33911

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.006117   0.007634    0.801   0.4245
x            1.436048   0.167126    8.593 2.41e-14 ***
v_hat_e      -1.158676   0.525710   -2.204   0.0293 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08771 on 129 degrees of freedom
Multiple R-squared:  0.365,    Adjusted R-squared:  0.3551
F-statistic: 37.07 on 2 and 129 DF,  p-value: 1.905e-13
```

$\therefore$ The P-value of $\hat{v}_e = 0.0293 > 0.01$ $\therefore$ 在 1% 顯著水準下, 我們無法拒絕市場報價是外生的假設。#

(g)

```
> #g小題
> model_iv_g<-ivreg(y~x|rank+pos,data=capm4)
> summary(model_iv_g)

Call:
ivreg(formula = y ~ x | rank + pos, data = capm4)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27725 -0.05091 -0.01043  0.04215  0.34665

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.006117   0.007763   0.788    0.432
x            1.436048   0.169934   8.451 5.06e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08918 on 130 degrees of freedom
Multiple R-Squared: 0.3384,    Adjusted R-squared: 0.3333
Wald test: 71.41 on 1 and 130 DF,  p-value: 5.059e-14
```

利用 IV 估計的 $\text{Beta} = 1.436048$ 大於利用 OLS 估計的 $\text{Beta} = 1.318947$,
如 (d) 小題之解釋, 此 IV 之估計亦符合我們的預期。 #

(h)

```
> cat("Sargan 統計量 (N*R2):", sargan_stat, "\n")
Sargan 統計量 (N*R2): 0.4157366
> cat("p-value:", p_val_sargan, "\n")
p-value: 0.5190721
```

H_0 : 工具變數與誤差項不相關 vs. H_1 : 工具變數與誤差項有相關

$\because P\text{-value} = 0.5190721 > 0.05 \therefore$ Do not reject H_0 at $\alpha = 0.05$

There is not enough evidence to say that the IV variances are relative to error term. #